

Name: Devansh Singh

PRN: 2070122072

Batch: A-2

Division: A

Subject: Programming with Java

ASSIGNMENT No : 8

TITLE : Write a Java program to demonstrate the use of the super keyword in inheritance.

Problem Statement

Write a Java program to demonstrate the use of the super keyword in inheritance.

Learning Objectives

To demonstrate the use of the super keyword for accessing parent class members.

Theory

The super keyword is used to refer to the immediate parent class in Java. It can invoke parent class constructors, access parent class variables, and call overridden methods. It ensures proper constructor chaining during object creation.

Output Screenshots

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp8/Main.java
Parent class method
Child class method
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Exercise

Q1. Create an Employee and Manager program where Manager accesses Employee details using super keyword.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp8/Exercise/Ex1.java
This is employee class
This is manager class
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Q2. Develop a Vehicle Insurance System where child insurance classes access parent vehicle information using the super keyword.

```
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$ java Exp8/Exercise/Ex2.java  
Brand : Toyota  
Model : Etios  
dslord@Fedora:~/Desktop/Coding Projects/Java/Java Lab$
```

Conclusion

This assignment helped in understanding the use of the super keyword in Java inheritance. It demonstrated how the super keyword is used to access parent class constructors, variables, and methods from a child class, ensuring proper initialization and reducing code duplication. The implementation improved my understanding of inheritance, constructor chaining, and method overriding in object-oriented programming. These concepts are widely used in developing desktop applications, web applications, Android applications, enterprise software, and other real-world Java programs that require efficient code reuse, maintainability, and hierarchical class design.